Ziqian Zhang

AI Research · LLM Evaluation · Reinforcement Learning

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Undergraduate researcher in LLM evaluation and RL; ICLR 2026 co-first author (equal contribution). Medical AI research and startup engineering experience.

Education

Lehigh University — Bethlehem, PA
Bachelor of Arts, double major in Computer Science and Mathematics — *Aug 2023 – Jan 2027*
GPA: 3.94/4.00 · Dean's List · **Expected graduation: January 2027**

Selected Publication

RankLLM: Weighted Ranking of LLMs by Quantifying Question Difficulty (ICLR 2026).
Ziqian Zhang*, Xingjian Hu*, et al. * Equal contribution. arxiv.org/abs/2602.12424v1

Research & Industry Experience

Red-flag Research, OpenLAIR — Bethlehem, PA
Project Lead — *Oct 2026 – Present*

Public screening for automated-research attacks on peer review openlair.github.io/red-flag-research.github.io

- Built a staged pipeline that checks submissions against earlier arXiv/OpenReview papers, aligns the closest passages, numbers and tables, and flags uncredited close matches for human review; screening all **43,013 ICLR 2027 submissions** in nine public stages.
- Released a benchmark of **20 anonymized real cases (34 target–source pairs)** with a human label for every case, standard-library evaluation tools and saved baseline outputs.

Xing Lab, Department of Radiation Oncology, Stanford Medicine — Stanford, CA
Student Research Intern (Prof. Lei Xing) — *Jun 2026 – Present*

- **WSI biomarker prediction & 3D CT segmentation:** Built interpretable feature-map-based MIL with strict TCGA→CPTAC external validation; improved external AUROC by **2+ percentage points** across biomarkers and MIL architectures. WSI manuscript submitted to *Nature Biomedical Engineering*.

LAIR Lab, Lehigh University — Bethlehem, PA
Undergraduate Research Assistant (Prof. Lichao Sun) — *May 2024 – May 2026*

RankLLM · Co-first author (equal contribution) github.com/Leozz04/EIP

- Derived bidirectional score propagation on a directed model–question bipartite graph; proved convergence to a unique stationary distribution using Perron–Frobenius and extended scoring to continuous partial credit. The non-parametric, linear-time method uses one damping parameter.
- Ran inference across **30 LLMs and 35,550 questions**; designed human-alignment, robustness, convergence, sensitivity and IRT evaluations. Achieved **90% human-difficulty agreement**, outperformed IRT baselines, converged in **0.006 s** on consumer hardware and maintained **ρ > 0.99** under perturbation. Mixed-scale pools reduced extreme misestimation by **83%**.

Medical-R1 · Co-author

- Designed structured rewards for four answer types (date, tuple, integer, decimal) across **70 clinical calculators** on MedCalc-Bench; trained **10 models (3B–14B)** with GRPO, RLOO and REINFORCE++ using customized VERL (FSDP + vLLM rollout).
- Completed **65+ training runs**, SFT→RL warm-start and no-reasoning ablations; localized critical self-attention/MLP components through layer-wise ablation and improved accuracy through model merging.

Entrepreneurship

Mygrants.ai — Bethlehem, PA
Co-founder & Engineer — *Oct 2025 – Present*

- Built a full-stack grant discovery and proposal platform: crawled and normalized faculty publications into researcher profiles with academic embeddings for semantic grant–faculty matching; generated and maintained COA (Collaborators & Other Affiliations) lists; integrated collaborative LaTeX editing. Reduced proposal preparation from weeks to hours.

Vibe-researcher (AI-Researcher + Research-Agent) — Bethlehem, PA
Founding Engineer — *Jan 2026 – Present*

- Built a multi-agent MVP converting natural-language research ideas into curated paper sets, structured innovation graphs and standardized benchmark instances; delivered reusable benchmark schemas and reproducible build scripts for faster evaluation loops.

Open-source Contribution

rubric-eval · OpenAI Agents SDK auto-capture — *Submitted / in review*
github.com/Kareem-Rashed/rubric-eval/pull/14

- Implemented a dependency-free Python adapter for SDK evaluation cases, preserving tool/search/handoff traces and distinguishing pending approval from execution. Added **7 focused regression tests**; recorded local full-suite validation: **38 passed, 1 skipped**. Hosted checks await maintainer authorization.